Q **August 6, 2026** **Alice Liu**

Three Coins Again

[12 pts]

A light vertical spring is fixed to a horizontal table at its lower end; its upper end carries a rigid horizontal tray of mass $M = 100\,\mathrm{g}$ (the tray's mass is *not* negligible). The spring constant is $k = 80\,\mathrm{N/m}$, and at its natural length the tray's upper surface is $L_0 = 0.25\,\mathrm{m}$ above the table.

Three identical coins, each of mass $m = 20\,\mathrm{g}$, are dropped one at a time from directly above the tray, each released from rest at a height $h = 0.45\,\mathrm{m}$ above the tray's upper surface.

The sequence is:

1. Coin 1 is released and collides *perfectly inelastically* with the tray (it sticks); the combined system then oscillates vertically. 2. When the system first returns to the tray's initial position, coin 2 is released and collides perfectly inelastically. 3. When the system first returns to the tray's initial position again, coin 3 is released and collides perfectly inelastically.

Take $g = 10\,\mathrm{m/s^2}$, neglect air resistance, assume the spring never exceeds its elastic limit, treat each collision as instantaneous, and neglect the impulse of gravity during each collision.

Assumptions

1. The spring is linear and remains within its elastic limit throughout.
2. The tray is rigid; the coins stick to it (and to each other) on impact.
3. Collisions are instantaneous; gravity's impulse during a collision is negligible.

Questions.

Q1. First coin. Just after coin 1 strikes the tray, find the combined system's velocity, and the amplitude of the subsequent oscillation.

Q2. Second coin. Just before coin 2 strikes, find the magnitude of the system's velocity.

Q3. Third coin. Just after coin 3 strikes, find the system's velocity.

Q4. Separation? After coin 3 strikes, during the upward motion, does coin 3 ever separate from the body below it? Justify your answer with a calculation.